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1 Future discussions

# More Kernel Optimizations for SGEMM

We have only scratched the surface of optimizing SGEMM. Other topics include:

1. **Memory coalescing:** arrange accesses so neighboring threads use neighboring addresses and require fewer memory transactions.
2. **Occupancy and register pressure:** balance per-thread resources against the number of warps available to hide latency.
3. **Shared-memory bank conflicts:** pad or rearrange tiles so threads in a warp do not serialize their accesses.
4. **Vectorized memory operations:** use aligned `float4` loads and stores to reduce instruction overhead.
5. **Warp tiling:** divide a block's output tile among warps to improve locality and register reuse.
6. **Autotuning:** search tile sizes and launch parameters for the target GPU instead of relying on one universal configuration.

# Alternative Programming Models and DSLs

In production, writing raw CUDA is rarely worth the effort.

1. Triton: tiling is at the heart of GPU optimization, so that becomes our new atomic unit instead of threads. Thread-level optimizations are done by a JIT compiler.
2. CUTLASS, CuTe DSL, Layout Algebra: Manually handling indexing is cumbersome and error-prone. We can compose common patterns with an algebra that can be checked at compile time.

# Kernel Fusion and Specialized Kernels

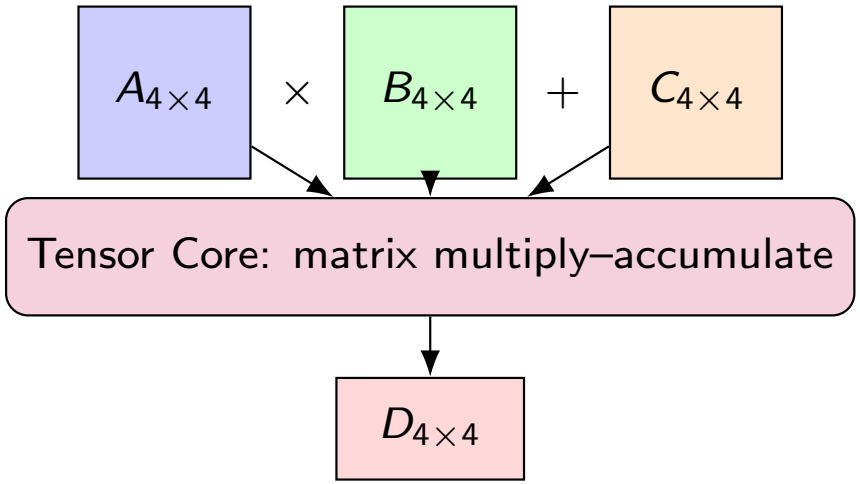
Kernels, be definition, have to read/write from DRAM. We can **fuse** kernels to increase performance.

1. Linear Layer + Nonlinearity = Fused MLP layer
2. FlashAttention series: complex fusion of the Attention Mechanism
3. Megakernels: taken to the extreme, we can fuse the operations necessary for the entire forward pass of a model into a **megakernel.**

# Continuing GPU evolution

## 2017: Volta Tensor Cores

- Tensor Cores perform mixed-precision matrix multiply–accumulate: $D = AB + C$.
- The original Volta unit multiplies FP16 inputs and accumulates in FP16 or FP32.
- Matrix multiplication becomes an explicit hardware primitive, alongside general-purpose CUDA cores.



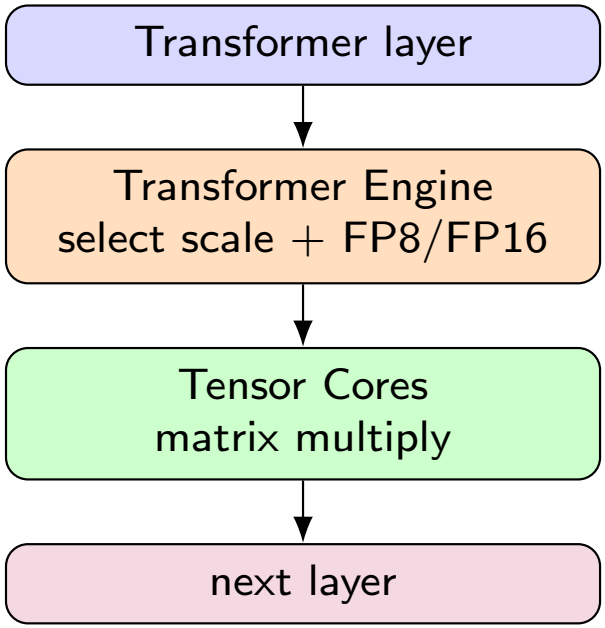

Operation shape shown for first-generation Volta Tensor Cores. Source:

NVIDIA, *Volta Architecture Whitepaper*, 2017.

# Continuing GPU evolution

**2020 onward: specialize around AI dataflow**

- Ampere (2020) adds TF32 and structured sparsity support.
- Hopper (2022) adds FP8 Tensor Cores and a Transformer Engine that dynamically chooses FP8 or FP16 per layer.
- Blackwell (2024) adds more low-precision formats and a second-generation Transformer Engine. This is more than incremental scaling.




Source: NVIDIA, *Hopper Architecture In-Depth* (2022).

# Continuing GPU evolution

## 2026: Vera Rubin, optimizing the rack

- The target is no longer an isolated GPU, but a complete rack-scale computing platform.
- Vera CPUs, Rubin GPUs, memory, NVLink, and networking are designed together to run one distributed AI workload.
- A high-bandwidth scale-up fabric lets many accelerators share model parameters and intermediate results.
- Performance now depends on communication, power, cooling, and software orchestration as well as matrix throughput.

**Unit of optimization**

Kernel

↓

GPU

↓

Server

↓

**Rack-scale system**

Source: NVIDIA, *Vera Rubin Architecture*, announced roadmap.

# Alternative Accelerator Architectures

## Big Picture

With respect to AI accelerators, GPUs are actually remarkably **general**. We can further specialize into AI-specific compute operations.

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3. NPUs: Serving AI workloads in power and heat constrained environments (mobile, consumer electronics, edge devices, etc.).

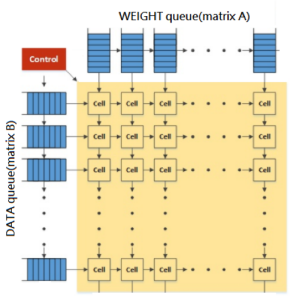
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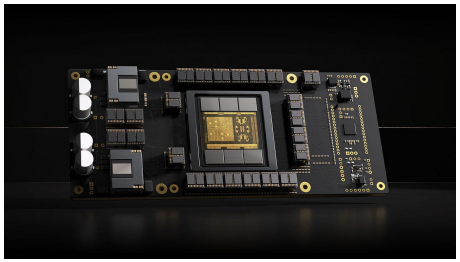
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3. NPUs: Serving AI workloads in power and heat constrained environments (mobile, consumer electronics, edge devices, etc.).
4. WSE (Wafer Scale Engine): Massive data bandwidth through multi-die wafers.

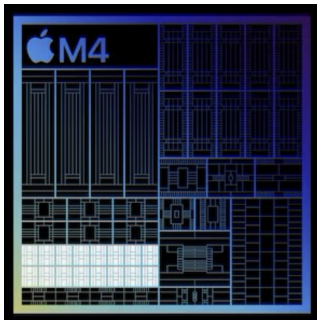
# Takeaways



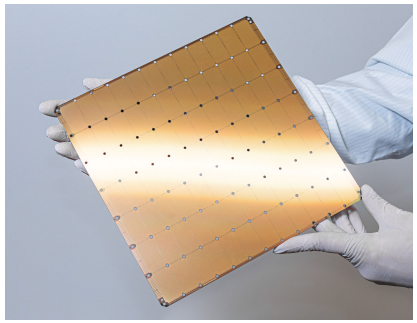

TPU systolic array



Etched Sohu




Apple M4 NPU



Cerebras WSE

**There is no universally best architecture**

Everything is a tradeoff. Optimizing for something always comes with a cost. **Workflow demands shape hardware evolution.**